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Progress in Perovskite Photocatalysis

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Semiconductor-assisted photocatalysis is a convenient approach to modulating chemical transformations with light. The choice of a semiconductor with desirable bandgap offers an interesting opportunity to tune the photo response as well as the energetics to induce reduction and oxidation processes. Semiconductor oxides (e.g., TiO_2, ZnO, and $BiVO_4$) and metal chalcogenides (e,g., CdS and CdSe) have reigned over the field of photocatalysis for the past few decades. With the emergence of metal halide perovskites, the focus has now shifted to their utilization as photocatalysts. The ease of preparation and ability to tune their bandgap by varying halide (Cl, Br, and I) composition makes them important candidates for photocatalytic applications. In our previous virtual issues we focused on the properties of perovskite nanocrystals and design of inorganic heterostructures.

Whereas metal halide perovskites exhibit impressive solar cell efficiencies, achieving close to the reported photoconversion efficiency of silicon solar cells, their application in photocatalysis is rather limited. The challenge to utilize them as photocatalysts comes from two major limitations: (1) chemical instability in polar solvents and (2) vulnerability of the surface to undergo transformation with chemical species present in solution. Recent research efforts have overcome some of these adverse scenarios and demonstrated ways to utilize metal halide perovskite nanocrystals and films in photocatalytic processes. The major developments in perovskite photocatalysis are summarized in recently published reviews.[1,2]

Surface modification with organic molecules/polymers or inorganic oxides/chalcogenides is an effective strategy to protect the semiconductor nanocrystals and remediate surface defects.[3−6] Thus, with a proper choice of surface modifier it is possible to improve the photocatalytic performance of perovskite nanocrystals. A recent success of capping $CsPbBr_3$ with ZnS has opened the door to carrying out photocatalytic experiments in polar solvents such as water.[6]

Basic understanding of interfacial charge-transfer processes is useful in identifying the factors that limit the photoconversion efficiency of photocatalytic process. By employing transient absorption spectroscopy it was possible to probe the selective electron and hole transfers from $CsPbBr_3$ nanoplatelets to adsorbed molecular acceptors such as benzoquinone (BQ) and phenothiazine (PTZ) (Figure 1).[7] Long-lived charge separation was also reported for $CsPbBr_3$−methyl viologen systems with the hole residing in $CsPbBr_3$ and the electron in the surface-bound methyl viologen moiety.[8] The ferrocenium/ferrocene (Fc^+/Fc^0) redox couple was employed to study the interfacial electron transfer to and from $CsPbBr_3$ NCs to a molecular redox mediator.[9] A lead-free Cs_2SnI_6 perovskite nanocrystal/SnS_2 nanosheet hybrid with a type II structure has proved to be effective in enhancing charge carrier lifetime and promoting the interfacial charge-transfer process.[10]

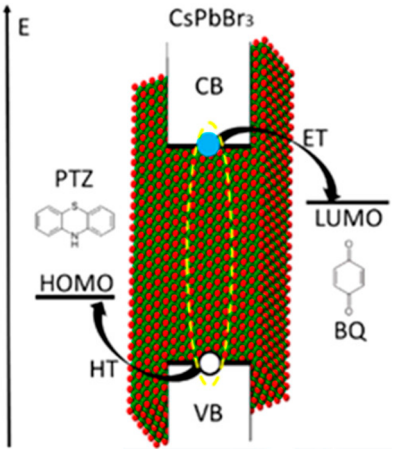

Figure 1. Schematic illustration of interfacial electron and hole transfer processes in $CsPbBr_3$ nanoplatelets. Reprinted from ref 7. Copyright 2019 American Chemical Society.

The photocatalytic application of perovskite nanocrystals has been explored in the transformation of organic compounds[11−15] as well as generation of solar fuels.[16−27] Of particular interest is their catalytic role in polymerization of 3,4-ethylenedioxythiophene,[11] C−H bond activation,[12] and α-alkylation of aldehydes.[13] In addition, these perovskite nanocrystals have also been found to be effective in the oxidation of benzylic alcohols[14] and degradation of organic compounds.[15] A major limitation of perovskite nanocrystals for hydrogen generation in aqueous solution is their chemical instability. However, use of HI and HBr maintained the stability of perovskite nanocrystals to generate hydrogen.[16,17] Another interesting approach to introduce stability is to grow a hybrid electron transport layer, a PCBM + TiO_2 film (18−40 nm thickness), over the perovskite film by atomic layer deposition (Figure 2)[18] or employ a tandem cell configuration.[19] CO_2 reduction using $CsPbBr_3$ nanocrystals has been carried out by depositing the nanocrystals on substrates

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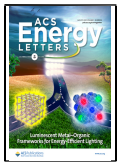

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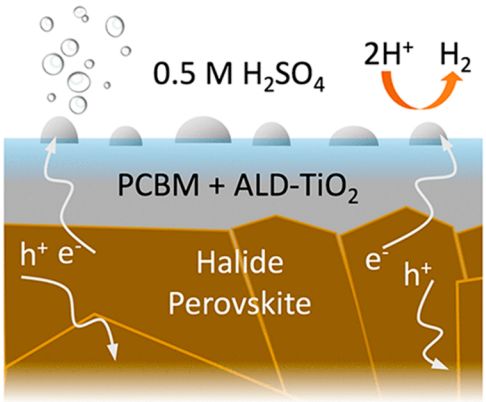

Figure 2. Schematic showing photoassisted proton reduction with halide perovskite photocathodes. Reprinted from ref 18. Copyright 2019 American Chemical Society.

such as graphene oxide,[20] MXene sheets,[21] and zeolitic imidazolate framework.[22] Perovskite nanocrystals in gaseous catalytic reaction[22] and as gas diffusion electrodes[23−27] have been employed to carry out visible light-induced photocatalytic reduction of CO_2.[22−27]

Perovskite photocatalysis is an emerging field, and we can expect some major developments in the coming months. This virtual issue, which highlights some of the early efforts to employ perovskite nanostructures in photocatalysis, should be useful for researchers to advance the goals of solar fuel production.

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